How long do you need?

Ground-Based Follow-up of LSST Transients — Hands-on Assignment

CASSA TDMMA Workshop 2026
Independent University, Bangladesh
Instrument: CRITO driving a simulated observatory
300 mm, 5.2 µm, INDI simulators

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Instructor copy. **This copy contains the worked solutions** (Section 5). To produce the student hand-out, define `\participantscopy`:

```
pdflatex -jobname=TDMMA-2026-participants \
  "\def\participantscopy{1}\input{TDMMA-2026-assignment.tex}"
```

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1 Setup

1.1 What you are flying

You will drive a complete observatory control system — CRITO — against *simulated* hardware. Every step is the real code path: the same INDI drivers, the same plate solver, the same exposure planner, the same archive writer. Only the photons are fictional.

Role	Driver you connect to	What it presents
Mount	INDI Telescope Simulator	300 mm optics, slews and tracks
Imaging camera	INDI CCD Simulator	5.2 µm pixels, star field from `gsc`
Guide camera	INDI CCD Simulator (guide role)	a second chip for PHD2
Focuser + wheel	INDI simulators	motorised focus, L/R/G/B/Ha/OIII/SII
Roof	INDI Dome Simulator	opens and closes
Broker feed	ALeRCE (ZTF, LSST-style)	live poll, or cached

Optical specification	
Focal length	300 mm
Pixel size	5.2 µm
Filters	L, R, G, B, Ha, OIII, SII
Sub-exposure cap	300 s

These are the values used both on paper and in the Exposure tab.

1.2 Participant setup (five minutes, at your seat)

1. Open an SSH tunnel to the workshop server. The instructor will give you an `SSH USER`, `SSH HOST`, and password during the session. Open Terminal (macOS/Linux) or PowerShell (Windows) and run:

```
ssh -N -L 8000:127.0.0.1:8000 SSH_USER@SSH_HOST
```

Replace `SSH_USER` and `SSH_HOST` with the values provided. The first connection may ask whether you trust the host key; confirm it only after the instructor verifies the fingerprint. Enter the session password when prompted. Password typing is intentionally invisible. Leave this terminal open for the entire exercise. You can stop the tunnel afterwards with `Ctrl+C`.

What the tunnel does. It maps port 8000 on your own computer to port 8000 on the workshop server. After it connects, open `http://localhost:8000` in your browser. The application is still running on the server; `localhost` now reaches it through the encrypted SSH connection. If local port 8000 is already occupied, use `-L 8080:127.0.0.1:8000` and open `http://localhost:8080` instead.

2. Claim your telescope.

1. Open `http://localhost:8000` through the tunnel above.
2. Log in with your workshop account: `workshopN` / `tdmma-2026`.
3. **Confirm the simulation badge** in the top bar. If it is not there, stop: you are pointed at real hardware and your account will (correctly) refuse to move anything.
4. On **Locations**, find your assigned telescope (`workshop1` → `Simulator 1`, and so on) and click **Operate**. Other assigned telescopes are disabled; an occupied one says **Busy**. Your claim is exclusive while you work.
5. **Devices** → **Scan** → **Auto-detect & connect all**. Six roles should bind: mount, camera, guide camera, focuser, filter wheel, roof.
6. **Exposure** tab → confirm a calibration table is loaded (gains 0/60/120/200; filters L, R, G, B, Ha, OIII, SII). If it says *no calibration table*, tell your instructor — see Section 1.3.

What your account can and cannot do. The `workshop` role is ranked with `viewer` and promoted to operator-level *only on a backend that declares itself a simulator*. So the same login flies your simulator and is a spectator on the real telescope. Reads — telemetry, archive, atlas, candidates — are never blocked, because watching the real rig work is half the point. A refused command names its reason:

```
{"detail":"this account is limited to a simulation backend
          - this one drives the real observatory"}   [403]
```

1.3 Instructor setup (before the session)

1. Bring up the simulated observatory.

```
docker compose -f deploy/docker-compose.yml --profile sim up -d --build
```

This starts one CRITO backend and five isolated INDI/PHD2 simulator containers. Everyone uses `http://<host>:8000`; the backend keeps all five telescope connections live and routes each authenticated workspace to its claimed rig.

The star field is not optional. INDI's CCD simulator draws its stars by shelling out to `gsc`, the Guide Star Catalogue tool. Without it *every frame is blank noise*, and blank frames cannot be plate-solved — so centring, autofocus and polar alignment all fail in ways that look like solver bugs and are not. The image compiles `gsc` from source for exactly this reason. Sanity check:

```
docker compose -f deploy/docker-compose.yml exec indi-sim-1 \
  gsc -c 83.822 -5.391 -r 30 -m 0 12 -n 5
```

Stars listed = fine. `gsc: not found` = wrong image.

2. Verify the simulated optics. The application image includes the ASTAP CLI and D05 catalogue. Compose sets all five simulators to 300 mm, matching `deploy/observatory.sim.yaml`. To verify simulator 1:

```
docker compose -f deploy/docker-compose.yml exec indi-sim-1 \
  indi_getprop "Telescope Simulator.TELESCOPE_INFO.*"
```

Focal length	Pixels	Field (1024 px)	ASTAP database
300 mm	5.2 µm	~1.0°	**D05**, ~100 MB

3. Backend environment.

```
# .env  (one shared backend)
CRITO_SOLVE_SCIENCE_FRAMES=true       # REQUIRED for Part 2: writes WCS into lights
CRITO_CALIBRATION_FILE=calibration/minicam8.example.yaml
CRITO_AUTH_SECRET=<one long random workshop secret>
CRITO_ALERCE_PROBABILITY=0.2          # loosen so the candidate list is never empty
CRITO_ALERCE_LOOKBACK_DAYS=14
```

Compose supplies the simulation flag, telescope endpoints, 300 mm/5.2 µm solver geometry, ASTAP path and D05 database path inside the containers. Do not put host macOS paths into these settings.

`CRITO_SOLVE_SCIENCE_FRAMES` is off by default. Light frames only carry `CRVAL1`/`CRVAL2` (the plate-solved sky coordinates of the frame centre) when this flag is on — it costs a solve per frame, so the default is `false`. Part 2 asks participants to read those keywords out of the header. Turn it on, or drop that item from the hand-in.

4. One backend, five telescope leases. The telescope pool is declared in `deploy/observatory.sim.yaml`. It maps `workshop1` to `sim-1`, through `workshop5` to `sim-5`; operators may use any simulator and an administrator can force-release one. The browser renews an active claim every 30 seconds. An abandoned claim expires after 120 seconds, while a second user trying to claim the same telescope gets HTTP 409 instead of sharing its controls.

5. Accounts.

```
docker compose -f deploy/docker-compose.yml exec crito-sim \
  python scripts/seed_workshop_users.py --count 5 --password 'tdmma-2026'
docker compose -f deploy/docker-compose.yml exec crito-sim \
  python scripts/seed_workshop_users.py --list
```

Idempotent — re-run after a database reset. New `workshop` accounts are **simulation-only by default**; the toggle to real hardware lives in the console's **Users** page, *Hardware* column, and is re-read from the database on every write (so revoking access takes effect immediately, not at the participant's next login).

1.4 Pre-flight checklist

	Check	If it fails
☐	SIMULATION badge in the top bar	wrong backend — do not proceed
☐	six devices connected after *Scan*	re-run auto-detect; check `indiserver` is up
☐	a test frame shows *stars*, not noise	`gsc` missing from the sim image
☐	Exposure tab lists gains and filters	`CRITO_CALIBRATION_FILE` unset
☐	Candidates tab has $\geq$ 1 target	lower `CRITO_ALERCE_PROBABILITY`, re-poll
☐	a plate solve succeeds	FOV hint vs. database size mismatch

2 The physics you need

Everything below is in **electrons**. This is the whole of the reference sheet; Part 1 needs nothing else.

2.1 Geometry

$$\text{plate scale:}\quad s = 206.265 \times \frac{p\,[\mu\text{m}]}{f\,[\text{mm}]} \quad [\text{arcsec/px}] \tag{1}$$

$$\text{PSF width:}\quad \theta_{\text{px}} = \frac{\text{seeing}\,[\text{arcsec}]}{s}, \qquad \sigma = \frac{\theta_{\text{px}}}{2.3548} \tag{2}$$

$$\text{aperture:}\quad n_{\text{pix}} = \pi\,(1.5\,\theta_{\text{px}})^2 \tag{3}$$

A radius of 1.5×FWHM encloses about 99 % of a Gaussian PSF's flux — that is the aperture the photometry is done in, and n_{pix} is how many pixels' worth of sky, dark and read noise you are forced to swallow along with the star.

2.2 Rates

$$\text{source:}\quad S = 10^{-0.4\,(m-\text{ZP})} \quad [\text{e}^-\,\text{s}^{-1}] \tag{4}$$

$$\text{PSF peak:}\quad P = \frac{S}{2\pi\sigma^2} \quad [\text{e}^-\,\text{s}^{-1}\,\text{px}^{-1}] \tag{5}$$

The zero point ZP is defined as the magnitude that yields exactly 1 $\text{e}^-\,\text{s}^{-1}$ through the whole system — optics, filter, sensor. It is measured per filter, on-sky, at your site; it is *not* a property of the camera.

2.3 The CCD equation

For a single sub-exposure of length t:

$$\text{SNR}_{\text{sub}} = \frac{S\,t}{\sqrt{\underbrace{S\,t}_{\text{source shot}} + n_{\text{pix}}\big(\underbrace{B\,t}_{\text{sky shot}} + \underbrace{D\,t}_{\text{dark}} + \underbrace{R^2}_{\text{read}}\big)}} \tag{6}$$

with B the sky rate ($\text{e}^-\,\text{s}^{-1}\,\text{px}^{-1}$), D the dark current ($\text{e}^-\,\text{s}^{-1}\,\text{px}^{-1}$) and R the read noise (e^-, *not* squared in the table — square it here).

2.4 The two bounds on a sub-exposure

$$\text{sky-limited floor:}\quad t \ge \frac{k\,R^2}{B}, \qquad k = 10 \tag{7}$$

$$\text{saturation ceiling:}\quad t \le \frac{0.7\,\text{FW}}{P_{\text{brightest}} + B + D} \tag{8}$$

Equation (7) says: make the sub long enough that sky shot variance is at least ten times the read variance, otherwise you pay the read-noise toll on every single frame you stack. Equation (8) says: the brightest star you care about must stay in the linear part of the well — 70 % of full well is the usual safe fraction. A third bound is operational, not physical: **your mount and guider cap the sub** regardless of what the photons allow.

2.5 Stacking

$$\mathrm{SNR}_{\mathrm{total}} = \mathrm{SNR}_{\mathrm{sub}}\sqrt{N}, \qquad N = \left\lceil \left(\frac{\mathrm{SNR}_{\mathrm{required}}}{\mathrm{SNR}_{\mathrm{sub}}}\right)^2 \right\rceil, \qquad t_{\mathrm{total}} = N\,t \tag{9}$$

$$\sigma_{\mathrm{mag}} = \frac{1.0857}{\mathrm{SNR}_{\mathrm{total}}} \tag{10}$$

2.6 The calibration table

These are the constants the planner is loaded with (`calibration/minicam8.example.yaml`). Read noise and full well change with gain, because on a CMOS sensor they must; sky rate and zero point are per *filter* and per *site*, and are gain-independent.

This exercise uses the gain-60 row and the R row. The rest is here so you can check that the table on screen matches, and so you can see that sky rate and zero point are properties of the filter and the site, not of the camera.

Gain setting	**Read noise** R [e$^-$]	**Full well** FW [e$^-$]	**System gain** [e$^-$/ADU]
0	5.0	51 000	0.80
60	3.2	38 000	0.45
120	1.1	13 000	0.16
200	0.9	9 000	0.10

Filter	**Sky** B [e$^-$ s^{-1} px^{-1}]	**Zero point** ZP
L	8.00	21.0
R	3.00	20.3
G	3.50	20.4
B	2.00	19.8
Ha	0.12	18.6
OIII	0.15	18.8
SII	0.10	18.3

Other	
Pixel pitch p	5.2 µm
Dark current D at −10 °C	0.01 e$^-$ s^{-1} px^{-1}
Focal length f	300 mm

3 The assignment

Part 1 — Plan it

25 minutes, on paper

The situation. It is 19:40 on a clear September evening in Dhaka. An alert comes through the broker: a supernova candidate discovered two nights ago, currently at $R = 20.0$ and still rising. The follow-up coordinator sends the same message to every node in the network:

> *"We need one R-band photometric point on this object tonight, good to about a tenth of a magnitude. Can you take it — and how long do you need?"*

You have to answer the second question with a number of minutes before you start. The object sets at 01:20 and three other targets are queued behind it, so an open-ended exposure is not an option, and a total that is too short returns a point too noisy to use.

Part 1 is that estimate: **how much telescope time does one photometric point on this star cost?** It is built from seven quantities, taken in order.

What you know when the alert arrives		**Where it comes from**
Target magnitude	$R = 20.0$	the broker's latest photometry
Required SNR	10	the coordinator asked for ∼0.1 mag; by Eq. (10), $\sigma_{\text{mag}} = 1.0857/\text{SNR}$
Filter	R	the network's light curves are in R
Gain setting	60	the rig's standard science gain
Sensor temperature	−10 °C	what the cooler holds on a September night
Seeing	3.0 arcsec FWHM	an ordinary night at this site
Focal length	300 mm, pixels 5.2 µm	the instrument you are connected to
Sub-exposure cap	**300 s**	past this the guiding drifts and stars trail

No console. No calculator app you cannot explain. Derive, *in this order*:

1.1 **Geometry:** the plate scale s, the FWHM in pixels, and the aperture size n_{pix}. (Not one of the seven numbers, but nothing else works without them. n_{pix} is how many pixels' worth of sky you must measure in order to measure one star.)

1.2 The **sky-limited floor** $t \geq 10R^2/B$ — the shortest sub worth taking. Below it you pay the read-noise toll on every frame in the stack.

1.3 The **saturation ceiling** $t \leq 0.7\,\text{FW}/(P + B + D)$, taking the target itself as the brightest thing in the aperture.

1.4 A **sub-exposure length** inside the window those two bounds leave. State the rule you used to choose it, and say *which* of the three limits — floor, saturation, guiding — binds.

1.5 SNR **per sub** from Equation (6). Write out all four noise terms separately and give each as a fraction of the total; that tells you what is limiting the observation.

1.6 The **number of subs** N and the **total on-sky time**. This is the figure you report back.

1.7 The achieved SNR and the resulting σ_{mag}. Compare it with the tenth of a magnitude that was requested.

▶ **Hand in:** seven numbers — floor, ceiling, sub length, SNR_{sub}, N, total time, σ_{mag} — *with the working.* A number without its working scores nothing. Then complete the reply: *"Yes — ______ minutes on sky, ____ subs of ____ s in R, photometry to ±____ mag."*

Estimate first. Before working through the arithmetic, write down how many minutes you expect a mag-20 star to need on this telescope for a 0.1 mag point. Compare it with your final answer. Most estimates are out by a factor of several, and the size and direction of the error is worth knowing.

Part 2 — Fly it

45 minutes, on the simulator

2.1 Log in to your `workshop` account and complete the pre-flight checklist.

2.2 Put Part 1 into the **Exposure** tab: focal length 300 mm, pixel size 5.2 µm, magnitude 20.0, required SNR 10, filter R, gain 60, temperature −10, seeing 3.0, max sub 300 s. **The**

planner must agree with your paper. If it does not, one of you is wrong — find out which, and write down what the disagreement was.

2.3 Go to **Candidates**, poll ALeRCE, and pick an observable target. Note its class, peak altitude, airmass and moon separation, and say in one line why it is or is not worth a slot tonight.

2.4 **Approve** → **Queue**, then **Observing** → **Launch**. Watch the whole chain and name each stage as it happens: slew → plate-solve → centre → autofocus → guide → expose → archive.

2.5 Open the archived frame and read its FITS header.

▶ **Hand in:** from the header — `OBJECT`, `FILTER`, `EXPTIME`, `GAIN`, `EGAIN`, `READNOIS`, `CRVAL1`, `CRVAL2`; and from the archive row — `HFR` and the star count. Plus one screenshot of the execution monitor mid-run.

Where each number lives. `OBJECT`, `FILTER`, `EXPTIME`, `GAIN`, `EGAIN` and `READNOIS` are written into the FITS header at the edge, when the frame is authored. `CRVAL1`/`CRVAL2` are injected *after* a successful plate solve, and only when `CRITO_SOLVE_SCIENCE_FRAMES=true`. HFR and star count are *not* header keywords at all: they are measured at ingest by ASTAP and stored on the archive row — read them in the console's Archive list.

Deliverable and marking

One page: **the numbers, the header, one screenshot**. Be ready to state, in two minutes, your total time and *which* of the three limits — sky floor, saturation, guiding — set your sub length.

Part	Marks	What earns them
1	55	Seven correct numbers with working (35); the completed reply (5); naming which bound binds the sub length, by how much, and why (15).
2	45	Planner reconciled with paper, any disagreement diagnosed (15); execution chain named stage by stage (10); header and archive values transcribed correctly (15); candidate justified in one line (5).

4 Hints

Read these only when you are stuck. They are ordered, and each gives away a little more than the last.

Part 1

H1.1. Do the geometry first and write down three numbers before you touch the CCD equation: s in arcsec/px, FWHM in px, and n_{pix}. Every later step uses n_{pix}, and it is the single easiest place to lose a factor of π.

H1.2. $206.265 \times 5.2/300$ is just under 3.6 arcsec/px, so at 3 arcsec seeing the PSF is narrower than one pixel. The instrument is under-sampled — that is a property of a wide-field rig, not an error — and it is why n_{pix} comes out at about five pixels.

Note also that $n_{\text{pix}} \propto \text{seeing}^2$, and that Equation (3) contains no camera property. The seeing, not the sensor, sets how many pixels of background are measured alongside the star. This matters again at step 1.5.

H1.3. For the ceiling you need the PSF *peak* rate, not the total: the star's light is spread over the PSF, and only the central pixel is at risk of saturating. That is Equation (5), and σ there is in *pixels*.

H1.4. The ceiling for a mag-20 star comes out at thousands of seconds. That is not an error. Consider whether saturation is the constraint that actually limits your exposure at this magnitude.

H1.5. A defensible rule for choosing the sub: *three times the sky-limited floor*, clamped into the window. At $3\times$ the floor, read noise is about 3 % of your noise budget — comfortably sky-limited, without making each sub so long that one satellite trail costs you a big fraction of the night.

H1.6. N must be a whole number, and you must round *up*. Rounding up means you overshoot the required SNR slightly — report the SNR you actually achieve, not the one you asked for.

Part 2

H2.1. If the planner disagrees with your paper by a large factor, check the focal length and pixel size in the form *first*: they must read 300 mm and 5.2 µm. Selecting a saved setup with different optics replaces both fields without warning, and every result downstream changes with them.

H2.2. If the planner disagrees by a few percent only, you are both right and the difference is rounding — in n_{pix}, or in where you rounded N. Say so explicitly; that is a correct answer, not a failure.

H2.3. Empty candidate list? The broker returned nothing, which is a real operational condition, not a bug. Lower `CRITO_ALERCE_PROBABILITY`, widen `CRITO_ALERCE_LOOKBACK_DAYS`, or clear the classifier and re-poll.

H2.4. Nothing launches? Check for a *manual override* badge in the Observing tab. Any manual mount or camera command pauses the queue by design — a human touching the telescope always wins. Clear the override and resume.

H2.5. To read the header without leaving the terminal:

```
curl -s localhost:8000/api/images/<id>/fits -o frame.fits
python -c "from astropy.io import fits; print(repr(fits.open('frame.fits')[0].header))"
```

5 Solutions

All values below were computed with `crito.transient.exposure` against `calibration/minicam8.example.yaml`, and agree with the console to the digits shown. Small differences in a participant's answer — a percent or two — are almost always rounding in n_{pix} or in N, and should be accepted.

5.1 Part 1, step by step

1.1 Geometry.

$$s = \frac{206.265 \times 5.2}{300} = \mathbf{3.575}\ \text{arcsec/px}$$
$$\theta_{\text{px}} = \frac{3.0}{3.575} = \mathbf{0.839}\ \text{px}, \qquad \sigma = \frac{0.839}{2.3548} = 0.356\ \text{px}$$
$$n_{\text{pix}} = \pi\,(1.5 \times 0.839)^2 = \pi \times 1.259^2 = \mathbf{4.98}\ \text{px}$$

The aperture is five pixels. The seeing disc is narrower than one pixel, so the instrument is under-sampled; the consequence is that measuring one star brings in only five pixels' worth of sky, dark and read noise.

Source rate.

$$S = 10^{-0.4\,(20.0-20.3)} = 10^{0.12} = \mathbf{1.318}\ \text{e}^-\ \text{s}^{-1}$$

1.2 Sky-limited floor.

$$t_{\text{min}} = \frac{10R^2}{B} = \frac{10 \times 3.2^2}{3.0} = \frac{102.4}{3.0} = \mathbf{34.1}\ \text{s}$$

Neither the target nor the optics enter this expression: the floor depends only on the read noise and the sky rate.

1.3 Saturation ceiling.

$$P = \frac{S}{2\pi\sigma^2} = \frac{1.318}{2\pi \times 0.356^2} = \frac{1.318}{0.798} = 1.652\ \text{e}^-\ \text{s}^{-1}\ \text{px}^{-1}$$

$$t_{\text{max}} = \frac{0.7 \times 38\,000}{1.652 + 3.0 + 0.01} = \frac{26\,600}{4.662} = \mathbf{5705}\ \text{s}$$

1.4 Sub-exposure length, and which bound binds. The window is $[34.1,\ 5705]$ s from the physics — but the guider caps the sub at 300 s, so the operative window is $[34.1,\ 300]$ s. Taking $3\times$ the floor:

$$t = 3 \times 34.13 = \mathbf{102.4}\ \text{s} \quad \in [34.1,\ 300]$$

The point of 1.4. The binding ceiling is the guiding cap, not saturation, by a factor of 19. At 20th magnitude on a sensor with a 38 ke$^-$ well, the star could integrate for an hour and a half before the central pixel filled; the limit is that the mount cannot track that long without trailing. Participants who report "saturation limits us to 300 s" have mis-read their own arithmetic. Those who quote the factor of 19 have read it correctly.

1.5 SNR per sub. At $t = 102.4$ s:

$$\begin{aligned}
\text{signal} &= S\,t = 1.318 \times 102.4 = 135.0\ \mathrm{e}^- \\
\text{source shot} &= 135.0 \\
\text{sky} &= n_{\text{pix}} B\,t = 4.98 \times 3.0 \times 102.4 = 1528.9 \\
\text{dark} &= n_{\text{pix}} D\,t = 4.98 \times 0.01 \times 102.4 = 5.1 \\
\text{read} &= n_{\text{pix}} R^2 = 4.98 \times 10.24 = 51.0
\end{aligned}$$

$$\text{total variance} = 1720.0\ \mathrm{e}^{-2}, \qquad \text{noise} = 41.5\ \mathrm{e}^-$$

$$\mathrm{SNR}_{\text{sub}} = \frac{135.0}{41.5} = \mathbf{3.26}$$

The frame is **sky-limited**: sky is 88.9 % of the variance, the source itself 7.8 %, read noise 3.0 % and dark 0.3 %.

Read noise at 3 %. Participants routinely ask why a quieter gain setting is not used. At gain 200 the read noise is 0.9 e$^-$ instead of 3.2, and it changes almost nothing, because read noise is already only 3.0 % of the variance. Remove it entirely and the SNR improves by $1/\sqrt{0.970} = 1.5\,\%$.

There is a structural reason. Choosing the sub as 3× the floor means $t = 3 \times 10R^2/B$, so $B\,t = 30R^2$: the sky term is thirty times the read term *by construction*, whatever R happens to be. A quieter sensor does not lower that fraction — it just lets you reach the same fraction with shorter subs. Read noise is a *noise* term, not a signal term; removing noise you have already made negligible cannot add photons. Only aperture, time, throughput or a darker sky do that.

Compare that with the effect of seeing. Sky dominates the budget, the sky term is $n_{\text{pix}} B\,t$, and $n_{\text{pix}} \propto \text{seeing}^2$. On a 2 arcsec night instead of a 3 arcsec one,

$$n_{\text{pix}} = \pi\,(1.5 \times 0.559)^2 = 2.21\ \text{px} \quad \text{instead of} \quad 4.98\ \text{px},$$

a factor of $(2/3)^2 = 0.44$. That lifts $\mathrm{SNR}_{\text{sub}}$ from 3.26 to 4.66, and the observation drops from 10 subs to **5** — 8.5 minutes instead of 17.1, exactly half.

Spend the effort on...	SNR **gain**	**in magnitudes**
a perfect sensor (read noise → 0)	1.015×	0.016
one arcsecond of seeing (3 " → 2 ")	1.43×	**0.39**

Seeing is worth 24 times more than the sensor here, at no cost. The reason is structural: seeing sets n_{pix}, and n_{pix} multiplies the term that dominates the noise budget. This is why site quality, focus and guiding are worth more attention than the camera spec sheet, and it is the single most transferable lesson in this assignment.

1.6 Number of subs, total time.

$$N = \left\lceil \left(\frac{10}{3.255}\right)^2 \right\rceil = \lceil 9.44 \rceil = \mathbf{10}, \qquad t_{\text{total}} = 10 \times 102.4 = 1024\ \text{s} = \mathbf{17.1}\ \text{min}$$

1.7 Achieved SNR and magnitude error.

$$\mathrm{SNR}_{\mathrm{total}} = 3.255\sqrt{10} = \mathbf{10.29}, \qquad \sigma_{\mathrm{mag}} = \frac{1.0857}{10.29} = \mathbf{0.106}\ \mathrm{mag}$$

Answer key — Part 1

Sky-limited floor	34.1 s
Saturation ceiling	5705 s (guiding cap 300 s binds instead)
Sub-exposure	102.4 s
SNR per sub	3.26
Number of subs N	10
Total on-sky time	17.1 min
σ_{mag}	0.106 mag

The reply they were asked for. *"Yes — 17 minutes on sky, 10 subs of 102 s in R, photometry to ±0.11 mag."* Most estimates are much longer, because mag 20 sounds beyond a small wide-field instrument; it is not, when the aperture contains only five pixels of sky. Comparing the estimates written down at the start against the 17 minutes is worth two minutes of discussion.

The second half of the reply matters as much. The star could have integrated for 5705 s before saturating, so the 102 s frame was set by the mount. On this instrument the limit on a single frame is mechanical, not photometric.

Common wrong answers, and what they mean.

- *Ceiling* ≈ 300 *s.* They took the cap as the physics. Ask them what would change if the guider were perfect.
- $n_{\mathrm{pix}} \approx 2.2$. They used $\pi\theta^2$ — a radius of one FWHM instead of 1.5.
- $N = 9.44$. Not rounded up. Then the achieved SNR is reported as exactly 10, which is a give-away.
- $\mathrm{SNR}_{\mathrm{sub}} \approx 6.3$. They forgot n_{pix} on the sky term — i.e. measured one pixel's worth of sky instead of the aperture's.
- *Read noise entered as* 10.24. R is squared *in the equation*; the table already gives R, not R^2.
- *A sub-pixel FWHM reported as an error.* It is correct: at 3.58 arcsec/px the seeing disc is narrower than one pixel. See H1.2.

5.2 Part 2, what a good answer looks like

2.2 Reconciliation. The Exposure tab should return, for the Part 1 inputs: plate scale 3.58 arcsec/px, FWHM 0.84 px, aperture 5 px, sub window ≥ 34.1 s / ≤ 300.0 s, recommended sub 102.4 s, SNR/sub 3.25, 10 subs, 17.1 min total, SNR 10.3, ±0.106 mag, *sky-limited.* An exact match is expected, because the console runs the same equations they just did by hand — that is the intended realisation.

The disagreement to look for is the optics. Selecting a saved setup with a different focal length and pixel size overwrites both fields without warning. Anyone whose total is out by a large factor should check those two boxes read 300 mm and 5.2 µm before looking elsewhere.

2.3 Candidate selection. Accept any target that clears 30° during astronomical dark with reasonable moon separation. A good answer notes airmass at peak (below ∼1.5 is comfortable), moon separation (a target 20° from a bright moon is a waste of a slot however high it is), and the class — an SN candidate is a better use of a follow-up node than a known AGN.

2.4 Naming the chain. All six stages, in order: slew, plate-solve, centre (the solve says where you *actually* are; the mount is corrected and re-solved), autofocus (HFR V-curve), guide (PHD2 calibrate then lock), expose $\times N$, archive. Participants who say "it slewed and then took pictures" have not watched it.

2.5 Header values.

Keyword	Expected	Where it comes from
`OBJECT`	the candidate's designation	the observation request
`FILTER`	`R` (or as planned)	filter wheel slot
`EXPTIME`	the sub length they planned	the plan, not the planner
`GAIN`	60	camera setting
`EGAIN`	0.45 e^-/ADU	calibration table, *measured*, not the setting
`READNOIS`	3.2 e^-	calibration table
`CRVAL1/2`	within arcminutes of the target RA/Dec	injected from the plate solve
HFR	a few px, stable across the run	ASTAP at ingest, on the archive row
star count	tens to hundreds	ASTAP at ingest, on the archive row

The teaching point: `EGAIN` and `READNOIS` in the header are what makes the frame reducible by *somebody else*, months later, without asking you anything. That is the difference between a picture and a data product.

5.3 Instructor's one-page answer card

Part	Quantity	Answer
1	plate scale / FWHM / n_{pix}	3.575 arcsec px^{-1} / 0.839 px / 4.98 px
1	source rate at $R = 20$	1.318 e^- s^{-1}
1	sky-limited floor	34.1 s
1	saturation ceiling	5705 s *(guiding cap 300 s binds)*
1	sub-exposure	102.4 s
1	SNR per sub	3.26
1	subs / total	10 / 17.1 min
1	achieved SNR / σ_{mag}	10.29 / 0.106 mag
1	the answer to the coordinator	*17 minutes*
1	noise budget share	sky 88.9 % / source 7.8 % / read 3.0 % / dark 0.3 %
1	sensor *vs.* seeing	+0.016 mag *vs.* +0.39 mag — seeing wins 24×
2	planner, same inputs	sub 102.4 s, 10 subs, 17.1 min, SNR 10.3, sky-limited
2	the disagreement to look for	a saved setup overwriting 300 mm / 5.2 µm

A Doing it from the command line

The Exposure tab, the CLI and the API all call the same function. Any of these is an acceptable way to do Part 2.

```
# the whole of Part 1, from the calibration table
python -m crito.transient.exposure \
  --calibration calibration/minicam8.example.yaml \
  --mag 20.0 --snr 10 --filter R --gain 60 --temp -10 \
  --focal-length 300 --pixel-size 5.2 --seeing 3 --max-sub 300

# same thing with the constants typed in by hand (no table needed)
python -m crito.transient.exposure \
  --mag 20.0 --snr 10 --read-noise 3.2 --full-well 38000 \
  --sky 3.0 --zp 20.3 --dark 0.01 \
  --focal-length 300 --pixel-size 5.2 --seeing 3 --max-sub 300

# protect a bright field star from saturation (extra credit)
python -m crito.transient.exposure ... --brightest-mag 12

# over HTTP
curl localhost:8000/api/tools/exposure/calibration
curl -X POST localhost:8000/api/tools/exposure \
  -H 'content-type: application/json' \
  -d '{"mag":20.0,"required_snr":10,"filter":"R","gain":60,
       "temp_c":-10,"seeing_arcsec":3,"max_sub_s":300}'

# the transient chain
curl localhost:8000/api/transient/night          # tonight's dark window
curl -X POST localhost:8000/api/transient/poll   # poll ALeRCE now
curl "localhost:8000/api/transient/candidates?group_by=class"
```

B Troubleshooting

Symptom	Cause / fix
`focal length unknown` (400)	no `focal_length_mm` in `observatory.yaml`; pass `-focal-length` or fill the form field
`calibration for gain N is incomplete`	`read_noise_e` or `full_well_e` is null in the table
`filter 'X' not in calibration`	add it under `filters:`, or tick *manual constants*
Exposure tab: *no calibration table*	`CRITO_CALIBRATION_FILE` unset or wrong path; restart the backend
Plan wants an absurd total time	sky-limited faint broadband target — that *is* the answer, not a bug: check n_{pix} and the sky rate
saturation ceiling below sky floor	bright star in a dark sky: lower the gain (more full well) or accept a read-noise penalty
Frames are blank noise	`gsc` missing from the simulator image; nothing will plate-solve
Nothing solves	FOV hint does not match the star database — the simulated optics are 300 mm/5.2 µm, which wants **D05**
No `CRVAL1`/`CRVAL2` in the header	`CRITO_SOLVE_SCIENCE_FRAMES` is `false` (the default)
Queue will not launch	*manual override* is latched — a manual command always pauses the queue; clear it and resume
403 on any move command	the account is `sim_only` and this backend is not a simulator; check the SIMULATION badge
Candidate list empty	the broker returned nothing: lower `CRITO_ALERCE_PROBABILITY`, widen the lookback, re-poll
Polar alignment always reads ∼0	the simulated mount is perfectly aligned; the maths is being exercised, not your judgement

C Symbols

Symbol	Meaning	Units
s	plate scale	arcsec/px
θ_{px}	seeing FWHM in pixels	px
σ	Gaussian PSF width, $\theta_{\text{px}}/2.3548$	px
n_{pix}	pixels in the photometry aperture	px
m	target magnitude	mag
ZP	zero point: the magnitude giving 1 $e^- s^{-1}$	mag
S	source count rate	$e^- s^{-1}$
P	PSF peak rate, central pixel	$e^- s^{-1} px^{-1}$
B	sky background rate	$e^- s^{-1} px^{-1}$
D	dark current	$e^- s^{-1} px^{-1}$
R	read noise (*not* squared)	e^-
FW	full well	e^-
t	sub-exposure length	s
N	number of subs	—
σ_{mag}	1-σ magnitude uncertainty	mag

Built on CRITO. The planner is `crito.transient.exposure`; the constants are `calibration/minicam8.example.yaml`. Replace them with your own measurements before you believe any of it.